Survey on Value-at-Risk and Recent Financial Crisis

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What is value-at-risk?

- Most widely used financial tool for risk management & required by Basel Capital Accord
- Definition: the amount that an organization can expect to lose less than x percent of the time (if positions are left unchanged for t days).
- Example: "Probability of 5 percent, horizon of 1 day, VaR of $1 million" means that the organization expects to suffer daily losses of $1 million or more on only 5 percent of business days.
- Keypoints: the market risk of a PORTFOLIO; a PROBABILITY measure; assuming a NORMAL market of volatility

Assumptions for conventional VaR models

- Parametric models (GARCH(1,1), RiskMetrics)
  - The specification of variance
  - The normality assumption of residuals

- Nonparametric models (Historical Simulation)
  - an underlying assumption is that the return distribution within the sample (in the whole time series) remains unchanged

Abandon VaR?

- Still appealing: one single number; applied to about any asset class; measure both individual risks and firm-wide risk.

- In recent years, computing power has increased substantially. Thus, estimating more sophisticated models induces only little higher computation costs. Maybe we can expect them to be common VaR estimation approaches.

Literature review

- Chen et al. (2012) Bayesian Forecasting for Financial Risk Management, Pre and Post the Global Financial Crisis
- GARCH models outperformed stochastic volatility models in almost all cases
- Asymmetric volatility models were clearly favored pre crisis, while at the 1% level during and post crisis, for a 1-day horizon, models with skewed-t errors ranked best, while integrated GARCH models were favored at the 5% level
- All models forecast VaR less accurately and anti-conservatively post crisis.

- McAleer et al. (2012) Has the Basel Accord Improved Risk Management During the Global Financial Crisis?
- The forecasts of VaR could be improved using a combination of models rather than a single model.
- An aggressive risk management strategy yielded the lowest mean capital charges, and had the highest frequency of minimizing daily capital charges, but also tended to violate too often. A conservative risk management strategy would have far fewer violations, and a correspondingly higher mean daily capital charges. The area between the bounds provided by the aggressive and conservative risk management strategies would seem to be a fertile area for future research.

Promising models

- Extreme Value Theory (EVT)
  - Extreme value distributions(EVDs) use generalized Pareto distribution to fit the extreme tail part of the data
  - Based on Balkema-de Haan-Pickandstheorem (Balkemaand de Haan1974)
- Conditional Autoregressive Value at Risk (CAViaR) + EVT
  - CAViaR was introduced by Engle and Manganelli (1999). The basic intuition is to model directly the evolution of the quantile over time, rather than the whole distribution of portfolio returns.
- Nonparametric Quantile Regression + EVT
- GARCH-copula + EVT

FURTHER

- Choose a promising one? ---Sophisticated models can be common VaR estimation approaches.
- Or form a combination?
- What are the criteria? --- Backtesting, Basel II criteria, Standard Deviation
- Survive the crisis?

thank you
danke
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gracias
tapadh leat
merci
obrigado
bedankt
dziękuję
sukriya
kop khun krap
go raibh maith agat
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grazie
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